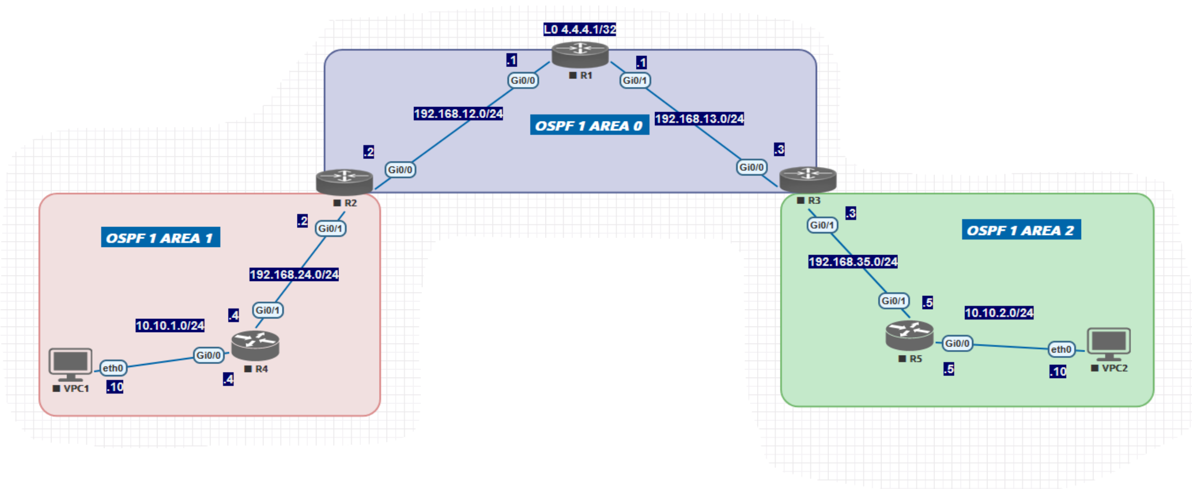
OSPFv2 Multi-Area Lab Manual

Areas, LSAs, Discontiguous Networks, Path Selection, Summarization & Route Filtering



Topology Reference Diagram

0. Topology Overview

This lab uses a five-router OSPFv2 topology divided into three OSPF areas. R1 is the backbone-only router and ABR-pair hub; R2 and R3 are Area Border Routers (ABRs) connecting Area 1 and Area 2, respectively, to the backbone (Area 0). R4 and R5 are internal routers in Area 1 and Area 2, each fronting an access-layer subnet with a virtual PC (VPC).

Device and Interface Addressing

Router	Interface	IP Address	Network	OSPF Area
R1	Gi0/0	192.168.12.1/24	192.168.12.0/24	Area 0
R1	Gi0/1	192.168.13.1/24	192.168.13.0/24	Area 0
R1	Lo0	4.4.4.1/32	4.4.4.1/32	Area 0
R2	Gi0/0	192.168.12.2/24	192.168.12.0/24	Area 0
R2	Gi0/1	192.168.24.2/24	192.168.24.0/24	Area 1
R3	Gi0/0	192.168.13.3/24	192.168.13.0/24	Area 0
R3	Gi0/1	192.168.35.3/24	192.168.35.0/24	Area 2
R4	Gi0/1	192.168.24.4/24	192.168.24.0/24	Area 1
R4	Gi0/0	10.10.1.1/25	10.10.1.0/25	Area 1
R4	Gi0/2	10.10.1.129/25	10.10.1.128/25	Area 1
R5	Gi0/1	192.168.35.5/24	192.168.35.0/24	Area 2
R5	Gi0/0	10.10.2.1/25	10.10.2.0/25	Area 2
R5	Gi0/2	10.10.2.129/25	10.10.2.128/25	Area 2
VPC1	eth0	10.10.1.10/25	10.10.1.0/25	n/a
VPC2	eth0	10.10.2.10/25	10.10.2.0/25	n/a
VPC3	eth0	10.10.1.130/25	10.10.1.128/25	n/a
VPC4	eth0	10.10.2.130/25	10.10.2.128/25	n/a

Note: R4 and R5 each front two LAN segments (a /25 pair per side) via Gi0/0 and Gi0/2. This split is intentional: it gives the Summarization task (Section 5) genuine sub-prefixes to consolidate. VPC1 and VPC2 remain on the first /25 of their respective pairs.

Area Assignment Summary

Area	Role	Routers / Links
Area 0 (Backbone)	Transit / backbone area	R1 - R2 link (192.168.12.0/24), R1 - R3 link (192.168.13.0/24), R1 Loopback 4.4.4.1/32
Area 1	Regular (non-backbone) area	R2 - R4 link (192.168.24.0/24), R4 LANs (10.10.1.0/25 and 10.10.1.128/25)
Area 2	Regular (non-backbone) area	R3 - R5 link (192.168.35.0/24), R5 LANs (10.10.2.0/25 and 10.10.2.128/25)

Note: All interfaces in this lab run OSPF process ID 1 ("router ospf 1"), as labelled in the diagram.

0.1 Initial Device Configurations

The configurations below represent the as-built baseline for all five routers before any of the lab tasks (Sections 1-6) are performed. Apply these configurations first, verify full adjacency, and then proceed to the numbered lab sections. Passwords, line configuration, and other platform defaults are omitted for brevity.

R1

```
hostname R1
!
interface Loopback0
 ip address 4.4.4.1 255.255.255.255
!
interface GigabitEthernet0/0
 ip address 192.168.12.1 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.13.1 255.255.255.0
 no shutdown
!
router ospf 1
 router-id 1.1.1.1
 network 4.4.4.1 0.0.0.0 area 0
 network 192.168.12.0 0.0.0.255 area 0
 network 192.168.13.0 0.0.0.255 area 0
```

R2 (ABR: Area 0 / Area 1)

```
hostname R2
!
interface GigabitEthernet0/0
 ip address 192.168.12.2 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.24.2 255.255.255.0
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 network 192.168.12.0 0.0.0.255 area 0
 network 192.168.24.0 0.0.0.255 area 1
```

R3 (ABR: Area 0 / Area 2)

```
hostname R3
!
interface GigabitEthernet0/0
 ip address 192.168.13.3 255.255.255.0
 no shutdown
```

```
!
interface GigabitEthernet0/1
 ip address 192.168.35.3 255.255.255.0
 no shutdown
!
router ospf 1
 router-id 3.3.3.3
 network 192.168.13.0 0.0.0.255 area 0
 network 192.168.35.0 0.0.0.255 area 2
```

R4 (internal Area 1 router)

R4 fronts two LAN segments (10.10.1.0/25 and 10.10.1.128/25) on separate interfaces. These two subnets are the targets of the summarization task in Section 5.

```
hostname R4
!
interface GigabitEthernet0/0
 ip address 10.10.1.1 255.255.255.128
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.24.4 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/2
 ip address 10.10.1.129 255.255.255.128
 no shutdown
!
router ospf 1
 router-id 4.4.4.4
 network 192.168.24.0 0.0.0.255 area 1
 network 10.10.1.0 0.0.0.127 area 1
 network 10.10.1.128 0.0.0.127 area 1
```

R5 (internal Area 2 router)

R5 fronts two LAN segments (10.10.2.0/25 and 10.10.2.128/25) on separate interfaces. These two subnets are the targets of the summarization task in Section 5.

```
hostname R5
!
interface GigabitEthernet0/0
 ip address 10.10.2.1 255.255.255.128
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.35.5 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/2
 ip address 10.10.2.129 255.255.255.128
 no shutdown
!
router ospf 1
 router-id 5.5.5.5
 network 192.168.35.0 0.0.0.255 area 2
```

```
 network 10.10.2.0 0.0.0.127 area 2
 network 10.10.2.128 0.0.0.127 area 2
```

VPC1 and VPC2

```
! VPC1 (connected to R4 Gi0/0)
ip address 10.10.1.10 255.255.255.128
default gateway 10.10.1.1

! VPC2 (connected to R5 Gi0/0)
ip address 10.10.2.10 255.255.255.128
default gateway 10.10.2.1
```

Baseline Verification

- show ip ospf neighbor on every router - all six links (R1-R2, R1-R3, R2-R4, R3-R5, plus R4/R5 LAN segments if multi-access) should show FULL state where an adjacency is expected
- show ip route ospf on R1 - should show O IA entries for 10.10.1.0/25, 10.10.1.128/25, 10.10.2.0/25, and 10.10.2.128/25 (four separate inter-area routes, prior to summarization)
- ping between VPC1 (10.10.1.10) and VPC2 (10.10.2.10) - should succeed end to end

1. OSPF Areas

OSPF scales by dividing a network into areas. Area 0 (the backbone) must exist in any multi-area design, and every other area must connect to Area 0 either directly or via a virtual link. Splitting the network into areas reduces the size of the link-state database (LSDB) on each router, limits the scope of SPF recalculations, and allows route summarization at area boundaries.

1.1 Objective

Configure R1-R5 so that the topology forms three OSPF areas as shown in the diagram: Area 0 (backbone), Area 1, and Area 2. R2 and R3 act as ABRs.

1.2 Configuration

On each router, enable OSPF process 1 and assign each interface (or network statement) to the correct area. Example for R1:

```
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 192.168.12.0 0.0.0.255 area 0
R1(config-router)# network 192.168.13.0 0.0.0.255 area 0
R1(config-router)# network 4.4.4.1 0.0.0.0 area 0
```

Example for R2 (ABR between Area 0 and Area 1):

```
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 192.168.12.0 0.0.0.255 area 0
R2(config-router)# network 192.168.24.0 0.0.0.255 area 1
```

Example for R3 (ABR between Area 0 and Area 2):

```
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 192.168.13.0 0.0.0.255 area 0
R3(config-router)# network 192.168.35.0 0.0.0.255 area 2
```

Example for R4 (internal Area 1 router) and R5 (internal Area 2 router). Note R4 and R5 each have two LAN subnets (a /25 pair) as described in Section 0.1:

```
R4(config)# router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 192.168.24.0 0.0.0.255 area 1
R4(config-router)# network 10.10.1.0 0.0.0.127 area 1
R4(config-router)# network 10.10.1.128 0.0.0.127 area 1

R5(config)# router ospf 1
R5(config-router)# router-id 5.5.5.5
R5(config-router)# network 192.168.35.0 0.0.0.255 area 2
R5(config-router)# network 10.10.2.0 0.0.0.127 area 2
R5(config-router)# network 10.10.2.128 0.0.0.127 area 2
```

1.3 Verification

- show ip ospf interface brief - confirm each interface is in the correct area
- show ip ospf neighbor - confirm full adjacencies form across all six links
- show ip ospf - confirm the router shows up as an ABR on R2 and R3 ("This router is an area border router")

2. Link-State Advertisements (LSAs)

OSPF builds its topology database from several LSA types, each flooded with a different scope. Understanding which LSA type generates which entry in the LSDB - and where that flooding stops - is essential to predicting OSPF behaviour across area boundaries.

2.1 LSA Types Used in This Topology

Type	Name	Originated By	Flooding Scope	Example in This Lab
1	Router LSA	Every router	Within its own area only	R1, R2, R3 each originate a Type 1 in Area 0; R2/R4 in Area 1; R3/R5 in Area 2
2	Network LSA	DR on a multi-access segment	Within its own area only	Generated for each /24 segment if a DR is elected (relevant on Ethernet links such as 192.168.12.0/24)
3	Summary (Inter-Area) LSA	ABR (R2, R3)	Injected into the adjacent area	R2 advertises 10.10.1.0/24 (Area 1) into Area 0 as a Type 3; R3 advertises 10.10.2.0/24 (Area 2) into Area 0 as a Type 3. R1 then re-advertises both into the opposite area

Type	Name	Originated By	Flooding Scope	Example in This Lab
4	ASBR Summary LSA	ABR	Adjacent areas	Not used unless redistribution/ASBR is introduced
5	AS External LSA	ASBR	Entire OSPF domain (Type 1-4 areas)	Generated only if external routes are redistributed into OSPF on any router

2.2 Demonstrating LSA Flooding Boundaries

Use the following commands on R1, R2, and R4 to observe how each LSA type appears in the database, and to confirm that Type 1/2 LSAs do not cross area boundaries while Type 3 LSAs do.

```
R2# show ip ospf database

             OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)
Link ID         ADV Router      Age    Seq#        Link count
1.1.1.1         1.1.1.1         ...    0x80000003 2
2.2.2.2         2.2.2.2         ...    0x80000003 1

                Router Link States (Area 1)
2.2.2.2         2.2.2.2         ...    0x80000002 1
4.4.4.4         4.4.4.4         ...    0x80000002 2

                Summary Net Link States (Area 0)
Link ID         ADV Router      Age    Seq#
10.10.2.0       3.3.3.3         ...    0x80000001

                Summary Net Link States (Area 1)
10.10.2.0       2.2.2.2         ...    0x80000001
```

Note that R2's Router LSA (Type 1) appears separately in both the Area 0 and Area 1 sections of its own database, because R2 is a member of both areas - this is the defining behaviour of an ABR. The 10.10.2.0/24 route (Area 2, owned by R5) reaches R2 only as a Type 3 Summary LSA, never as a Type 1 or Type 2.

2.3 Verification Commands

- show ip ospf database - full LSDB broken out by area and LSA type
- show ip ospf database router - Type 1 LSAs only
- show ip ospf database network - Type 2 LSAs only
- show ip ospf database summary - Type 3 LSAs only
- show ip route ospf - confirm O IA (inter-area) routes correspond to received Type 3 LSAs

3. Discontiguous Networks

A discontiguous network exists when two portions of the same OSPF area, or two areas that should be contiguous through the backbone, are physically separated by another area or by a non-OSPF link. The most common case in OSPF is a discontiguous backbone: a non-backbone area's ABR does not have a direct link to Area 0, so Area 0 itself is split into disconnected pieces. OSPF solves this with a virtual link, which logically extends Area 0 through a transit area.

3.1 Scenario

In the baseline topology, R2 and R3 both connect directly to R1 in Area 0, so the backbone is contiguous and no special configuration is required. To practise discontiguous-network recovery, introduce the following fault and repair it:

1. On R3, remove the Gi0/0 link to R1 from Area 0 (simulating a topology where R3 has no direct backbone connection).
2. Observe that R3 and R5 (Area 2) lose all inter-area routes, because Area 2 is now isolated from Area 0 - the backbone is discontiguous.
3. Add a second physical link between R1 and R3 and place it into a new non-backbone area, Area 3, on both routers.
4. Repair backbone contiguity by configuring a virtual link between R1 and R3, using Area 3 as the transit area.

3.2 Fault Injection (R3)

```
R3(config)# router ospf 1
R3(config-router)# no network 192.168.13.0 0.0.0.255 area 0
```

Note: After this change, run 'show ip ospf interface brief' on R3 - Gi0/0 (192.168.13.0/24) is no longer in any OSPF area, and R3 no longer has any interface in Area 0. 'show ip route' on R4/VPC1 will no longer show 10.10.2.0/24 (or 10.10.2.0/25 + 10.10.2.128/25) or 192.168.35.0/24.

3.3 Establish a Transit Area (Area 3)

A virtual link requires both ABR endpoints to share a common non-backbone area with a real intra-area path between them - the virtual-link configuration alone does not create that path. Add a new link between R1 and R3 (for example, R1 Gi0/2 to R3 Gi0/2) and place it into Area 3 on both routers:

```
! On R1
R1(config)# interface GigabitEthernet0/2
R1(config-if)# ip address 192.168.130.1 255.255.255.0
R1(config-if)# no shutdown
R1(config)# router ospf 1
R1(config-router)# network 192.168.130.0 0.0.0.255 area 3

! On R3
R3(config)# interface GigabitEthernet0/2
R3(config-if)# ip address 192.168.130.3 255.255.255.0
R3(config-if)# no shutdown
R3(config)# router ospf 1
R3(config-router)# network 192.168.130.0 0.0.0.255 area 3
```

Verify with 'show ip ospf neighbor' that R1 and R3 form a FULL adjacency over this new Area 3 link, and with 'show ip route' that each router has an intra-area (O) route to the other's Area 3 interface. This intra-area path through Area 3 is what the virtual link will ride on.

3.4 Repair with a Virtual Link

A virtual link is configured between the two ABRs that need to be logically connected to the backbone, using the transit area's Router ID of the remote endpoint. The transit area in the 'area <N> virtual-link <RID>' command must match the area that actually connects both endpoints - here, Area 3.

```
R1(config)# router ospf 1
R1(config-router)# area 3 virtual-link 3.3.3.3

R3(config)# router ospf 1
R3(config-router)# area 3 virtual-link 1.1.1.1
```

This creates a logical point-to-point Area 0 link between R1 and R3, transported across Area 3. R3 now has a virtual presence in Area 0, restoring contiguity for the backbone and re-enabling inter-area routing to and from Area 2.

Note: If the virtual link was instead configured as 'area 2 virtual-link ...' while the transit link sits in Area 3, 'show ip ospf virtual-links' will show the link as Down with a cost of 65535 (OSPF's infinite metric). This happens because R1 has no presence in Area 2 at all, so it cannot compute any intra-area cost to R3 through that area. The area number in the virtual-link command must always match the area that physically/logically connects both ABRs - in this design, Area 3.

3.5 Requirements and Caveats

- The transit area (Area 3 in this design) must not be a stub area.
- Both endpoints must reference each other's OSPF Router ID, not an IP address.
- The transit area number used in 'area <N> virtual-link ...' must match an area where both ABRs have an interface and a real intra-area path to each other - mismatched area numbers result in a virtual link stuck Down with cost 65535.
- Virtual links are a migration/repair tool, not a long-term design choice - prefer redesigning physical topology so every area touches Area 0 directly.

3.6 Verification

- show ip ospf virtual-links - confirms the virtual link's transit area is 3, the state is "Point-to-Point"/Up, and the cost is a real (non-65535) value
- show ip ospf neighbor - a virtual-link neighbor appears with interface type "Virtual Link" in FULL state
- show ip route - confirm 192.168.35.0/24 and 10.10.2.0/25 + 10.10.2.128/25 (or the summarized 10.10.2.0/24, if Section 5 has been completed) reappear on R2/R4 as O IA routes
- debug ip ospf adj - if the virtual link remains Down after the area numbers are corrected, use this to confirm Hello/DBD exchange across the Area 3 link (checks for MTU or authentication mismatches)

4. OSPF Path Selection

OSPF chooses the best path using cost, which by default is derived from interface bandwidth (cost = reference bandwidth / interface bandwidth, default reference bandwidth 100,000 = 100 Mbps). When multiple paths to the same destination exist, OSPF also applies a strict route-type preference order before comparing cost.

4.1 Route-Type Preference Order

5. Intra-area (O) - learned via Type 1/2 LSAs from within the same area
6. Inter-area (O IA) - learned via Type 3 Summary LSAs from another area
7. External Type 1 (O E1) - cost = external cost + internal cost to the ASBR
8. External Type 2 (O E2) - cost = external cost only (default for redistribution)

A lower-numbered category always wins, even if its accumulated cost is numerically higher than a route in a lower-preference category.

4.2 Cost Calculation Example

Assume default GigabitEthernet cost of 1 on every link in the diagram (reference bandwidth 100,000 Kbps / 1,000,000 Kbps, rounded up to 1). The OSPF cost from VPC1's subnet (10.10.1.0/25, owned by R4) to VPC2's subnet (10.10.2.0/25, owned by R5) is the sum of outgoing interface costs along the path:

Hop	Outgoing Interface	Cost Added	Running Total
R4 -> R2	Gi0/1 (192.168.24.0/24)	1	1
R2 -> R1	Gi0/0 (192.168.12.0/24)	1	2
R1 -> R3	Gi0/1 (192.168.13.0/24)	1	3
R3 -> R5	Gi0/1 (192.168.35.0/24)	1	4

Total OSPF cost from R4 to 10.10.2.0/24 = 4. Because R2-R1-R3 is the only inter-area path in this topology, this is also the only route - but the lab below introduces a second path so cost-based selection can be observed directly.

4.3 Lab: Influence Path Selection with Cost

To make path selection observable, temporarily add a second link directly between R2 and R3 (e.g. a new GigabitEthernet segment 192.168.23.0/24, both interfaces placed in Area 0). With this link in place, R2 has two paths toward 10.10.2.0/24: via R1 (cost 3) or directly via R3 (cost 2). Confirm the lower-cost direct path is preferred, then raise its cost to force traffic back over R1.

Note: Changing ospf cost on R2 Gi0/2 interface only changes R2's routing table only.
R3's nexthop for prefixes 10.1.1.0/25,10.1.1.12/25 and 192.168.24.0/24 remains via new direct link R2-R3. And for prefix 192.168.12.0/24, R3 installs 2 routes via R1 and R2 with equal cost of 2

```
! On R2, raise the cost of the direct R2-R3 link above the R1 path
R2(config)# interface gi0/2
R2(config-if)# ip ospf cost 100

! Equivalent change can be made on R3's matching interface
R3(config)# interface gi0/2
R3(config-if)# ip ospf cost 100
```

4.4 Verification

- show ip route ospf - inspect the metric (cost) shown for each O/O IA entry
- show ip ospf interface gi0/0 - confirm the configured/derived cost per interface
- traceroute 10.10.2.10 - confirm the actual forwarding path matches the lowest-cost OSPF path
- show ip ospf border-routers - view ABR/ASBR reachability and the intra-area cost used to reach each ABR

5. Route Summarization

Summarization reduces the number of routes advertised between areas (inter-area summarization on an ABR) or between OSPF and other domains (external summarization on an ASBR). In this topology, R2 and R3 are ideal summarization points because each sits between the backbone and a single access subnet.

5.1 Objective

As established in the baseline configuration (Section 0.1), each of R4 and R5 fronts two contiguous /25 LAN segments rather than a single /24:

- Area 1: 10.10.1.0/25 and 10.10.1.128/25, both summarizable as 10.10.1.0/24
- Area 2: 10.10.2.0/25 and 10.10.2.128/25, both summarizable as 10.10.2.0/24

Before summarization, R1 receives four separate Type 3 Summary LSAs (one per /25). The objective of this task is to consolidate each pair into a single /24 summary at the ABR before it is advertised into Area 0.

5.2 Inter-Area Summarization on the ABRs

Inter-area summarization is configured on the ABR with the 'area range' command, under the area that contains the more-specific routes. Set cost explicitly. Doing show prevents frequent Type 3 LSA generation.

```
! On R2 (ABR for Area 1) - summarize Area 1 routes before advertising into Area 0
R2(config)# router ospf 1
R2(config-router)# area 1 range 10.10.1.0 255.255.255.0 cost 2

! On R3 (ABR for Area 2) - summarize Area 2 routes before advertising into Area 0
R3(config)# router ospf 1
R3(config-router)# area 2 range 10.10.2.0 255.255.255.0 cost 2
```

After this change, R1 (and R3, for Area-1 routes / R2, for Area-2 routes) will see a single Type 3 LSA for 10.10.1.0/24 and a single Type 3 LSA for 10.10.2.0/24, instead of two /25 entries each.

5.3 Effect on the LSDB and Routing Table

Before Summarization	After Summarization
R1 LSDB (Area 0) contains Type 3 entries for 10.10.1.0/25, 10.10.1.128/25, 10.10.2.0/25, 10.10.2.128/25	R1 LSDB (Area 0) contains a single Type 3 entry for 10.10.1.0/24 and a single Type 3 entry for 10.10.2.0/24

Before Summarization	After Summarization
A flap of either /25 in Area 1 triggers a new Type 3 LSA flood into Area 0 and Area 2	A flap of either /25 inside Area 1 is hidden from Area 0/Area 2 as long as the summary range remains active

5.4 Verification

- show ip ospf database summary - confirm only the summarized /24 entries are present
- show ip route ospf on R1, R4, R5 - confirm a single O IA 10.10.1.0/24 and O IA 10.10.2.0/24 entry each
- show ip ospf - the 'area range' configuration is echoed under the relevant area

Note: If no more-specific routes currently exist under an 'area range', the ABR will not generate the summary LSA at all - the range command only activates once at least one contributing subnet is present in the LSDB.

6. Route Filtering

Route filtering controls which routes are installed into the routing table or advertised to neighbours. Because OSPF Type 3 LSAs propagate the existence of inter-area routes via flooding (not via a distance-vector update process), a standard 'distribute-list ... in' on a non-ABR router filters only the local RIB - it does not stop the LSA from being flooded. To actually prevent a route from being advertised between areas, filtering must be applied on the ABR using an LSA Type 3 filter ('area filter-list').

6.1 Objective

Allow the 10.10.2.0/24 summary (Area 2, behind R5, summarized at R3 per Section 5) to be advertised normally into Area 0, but prevent it from being advertised onward into Area 1. Routers R1 and R3 should retain a route to 10.10.2.0/24, while R4 and VPC1 (Area 1) should not. 192.168.35.0/24 and all other Area 2/Area 0 routes should continue to reach Area 1 normally.

Note: This is a different goal from 'area range ... not-advertise' on R3, which would suppress 10.10.2.0/24 everywhere (including Area 0 and R1). Here the route is intentionally permitted into Area 0 but selectively withheld from a specific other area - this requires a Type 3 LSA filter at the ABR on the receiving side of that area, R2, not a change to the summarization at R3.

6.2 Filtering Type 3 LSAs at the ABR

On R2 (the ABR for Area 1), define a prefix list that denies 10.10.2.0/24 and permits everything else, then apply it on the Area 1 boundary with 'area filter-list', using direction 'in' to filter Type 3 LSAs as they are received into Area 1 from the backbone.

```
R2(config)# ip prefix-list AREA1-IN seq 10 deny 10.10.2.0/24
R2(config)# ip prefix-list AREA1-IN seq 20 permit 0.0.0.0/0 le 32

R2(config)# router ospf 1
R2(config-router)# area 1 filter-list prefix AREA1-IN in
```

With this filter applied, R2 still receives the 10.10.2.0/24 Type 3 LSA from R1 in Area 0 and keeps it in its Area 0 LSDB, but does not re-originate it as a Type 3 LSA into Area 1. R4 and

VPC1 therefore have no route to 10.10.2.0/24, while R1, R2, and R3 retain it, and 192.168.35.0/24 (and any other Area 0/Area 2 routes matching the permit line) continue to reach Area 1 normally.

Note: Because the prefix-list entry 'deny 10.10.2.0/24' matches the exact /24 summary advertised by R3, no 'le'/'ge' modifier is needed here - this works cleanly precisely because R2 is filtering the already-summarized /24, not the component /25s. Had summarization (Section 5) not been performed, R2 would need separate 'deny 10.10.2.0/25' and 'deny 10.10.2.128/25' entries to match each component LSA individually.

Note: The area number in 'area <id> filter-list' refers to the area on the side of the ABR where the filter is applied - here, Area 1, the area being protected from the prefix. Direction 'in' filters Type 3 LSAs being received into that area from elsewhere (Area 0); direction 'out' filters Type 3 LSAs being sent out of that area to the rest of the OSPF domain.

6.3 Filtering the Local RIB with a Distribute List

Section 6.2's 'area 1 filter-list' withholds the Type 3 LSA for 10.10.2.0/24 from Area 1 entirely - once that filter is active, no router in Area 1 (including R4) has an LSA for this prefix to act on, so a local filter on R4 would have nothing to suppress. To contrast the two mechanisms, remove the area filter-list from R2 before continuing:

```
R2(config)# router ospf 1
R2(config-router)# no area 1 filter-list prefix AREA1-IN in
```

With the area filter-list removed, R2 again re-originates a Type 3 LSA for 10.10.2.0/24 into Area 1, and R4 receives it normally. Now apply a 'distribute-list' on R4 instead, to keep 10.10.2.0/24 out of R4's routing table only:

```
R4(config)# ip prefix-list BLOCK-102 seq 10 deny 10.10.2.0/24
R4(config)# ip prefix-list BLOCK-102 seq 20 permit 0.0.0.0/0 le 32

R4(config)# router ospf 1
R4(config-router)# distribute-list prefix BLOCK-102 in
```

After this change, R4 stops installing 10.10.2.0/24 in its routing table, but its LSDB is unaffected - R4 still receives and stores the Type 3 LSA, and would still forward it on to any further routers behind it in Area 1 (none exist in this topology, but the principle holds for a larger Area 1). This is the key difference from 6.2: the distribute-list hides the route from one router's RIB without affecting what is flooded through the area.

Note: In summary: 'area 1 filter-list ... in' on R2 stops the LSA from entering Area 1 at all - no router in Area 1 can see or use the route. 'distribute-list ... in' on R4 still lets the LSA into Area 1 and into R4's LSDB, but R4 alone declines to install it in its RIB. The two are not meant to be combined for the same prefix on the same path - 6.2 demonstrates area-wide filtering at the ABR, and 6.3 demonstrates router-local RIB filtering, as alternatives to each other.

6.4 Verification

- With 6.2's area filter-list active: show ip route 10.10.2.0 - confirm the route is present on R1/R2/R3 but absent on R4; show ip ospf database summary 10.10.2.0 on R2 - confirm the LSA is present in the Area 0 database but not re-originated into the Area 1 database; on R4, confirm the LSA is absent entirely

- With the area filter-list removed and 6.3's distribute-list active on R4 instead: show ip ospf database summary 10.10.2.0 on R4 - confirm the LSA IS present (received and stored); show ip route 10.10.2.0 on R4 - confirm the route is NOT installed
- show ip prefix-list detail - confirm hit counts on each sequence to verify the filter is matching traffic
- ping 10.10.2.10 from VPC1 - confirm reachability fails whenever 10.10.2.0/24 is absent from R4's routing table, under either filtering method

Review Questions

Answer the following questions based on the topology, configurations, and observations from this lab. Space is provided for written answers.

Areas

1. Which routers in this topology are Area Border Routers (ABRs), and which areas does each one border?

---

2. Why must every non-backbone area in an OSPF design connect to Area 0?

---

3. What command(s) would you use to confirm that R4's interfaces are correctly assigned to Area 1?

Link-State Advertisements (LSAs)

4. What is the difference in flooding scope between a Type 1 (Router) LSA and a Type 3 (Summary) LSA?

---

5. On R2, the Router LSA appears in both the Area 0 and Area 1 sections of the LSDB. Explain why.

6. How does 10.10.2.0/24 (owned by R5 in Area 2) appear in R4's LSDB - as which LSA type, and originated by which router?

---

7. Under what circumstance would a Type 5 LSA appear in this OSPF domain?

Discontiguous Networks

8. After R3's Gi0/0 link to R1 was removed from Area 0 in the fault-injection exercise, why did Area 2 lose all inter-area routes?

---

9. Explain, in your own words, what a virtual link does and why a new Area 3 link between R1 and R3 was needed before the virtual link could come up.

---

10. What two pieces of information must match between the R1 and R3 'area 3 virtual-link' configurations for the link to come up?

---

11. Suppose the virtual link is configured as 'area 2 virtual-link ...' but the new R1-R3 link is placed in Area 3. What would 'show ip ospf virtual-links' show, and why?

---

12. Why can a transit area for a virtual link not itself be configured as a stub area?

OSPF Path Selection

13. List the four OSPF route-type categories in order of preference, from most to least preferred.

---

14. Calculate the OSPF cost from R5 to 10.10.1.0/24, assuming default GigabitEthernet cost of 1 on every link. Show your working.

---

15. In the cost-manipulation lab, after raising the R2-R3 direct link cost to 100, which path does traffic from R2 to 10.10.2.0/24 take, and why?

---

16. Would an O E1 route with cost 50 ever lose to an O IA route with cost 200? Explain why or why not.

Summarization

17. Why was the 'area range' command configured under Area 1 on R2 rather than under Area 0?

---

18. After summarization, what happens to R1's LSDB entries for 10.10.1.0/25 and 10.10.1.128/25?

---

19. Describe one operational benefit of summarizing 10.10.1.0/25 and 10.10.1.128/25 into a single 10.10.1.0/24 route at R2.

---

20. What happens if you configure 'area 1 range 10.10.1.0 255.255.255.0' on R2 before any matching subnet exists in Area 1's LSDB?

Route Filtering

21. Why is 'area range ... not-advertise' on R3 not the right tool for this task, given that R1 and R3 must still retain a route to 10.10.2.0/24?

---

22. Explain the difference in effect between an 'area 1 filter-list ... in' on R2 and a 'distribute-list ... in' on R4, both targeting 10.10.2.0/24. Why must the area filter-list be removed before the distribute-list's effect can be observed?

---

23. After applying 'area 1 filter-list prefix AREA1-IN in' on R2, what would you expect to see (or not see) in R2's 'show ip ospf database summary' output for 10.10.2.0/24, for the Area 0 database versus the Area 1 database?

---

24. With the area filter-list removed and a distribute-list applied on R4 instead, why does R4's LSDB still contain the Type 3 LSA for 10.10.2.0/24 even though 'show ip route' on R4 does not show the route?

---

25. Propose a real-world scenario (other than this lab) where filtering a Type 3 LSA at an ABR's 'in' direction toward a specific area would be operationally useful.

Command	Purpose
show ip ospf neighbor	Verify adjacency state on every link
show ip ospf interface brief	Confirm area assignment and cost per interface
show ip ospf database	View the full LSDB, organized by area and LSA type
show ip route ospf	View installed OSPF routes with type (O, O IA, O E1/E2) and metric
show ip ospf border-routers	List known ABRs/ASBRs and the intra-area cost to reach them
show ip ospf virtual-links	Check status of any configured virtual links
show ip prefix-list detail	Inspect prefix-list match counters used in filtering